Sri Lanka Institute of Information Technology (SLIIT)

IT3021 – Data Warehousing and Business Intelligence
Assignment 2

Student Name: Ahamed MMU

Student ID: IT23692978

Degree Program: BSc (Hons) in Information Technology Specializing in Data Science

Submission Date: 02.04.2026

Contents

1 Data source....3
1.1 Data Source....3
1.2Entities and Their Purpose....3
2 SSAS Cube implementation....5
2.1 Data Loading into Data Source....5
2.2 Creating Data Source View (DSV)....6
2.3 Cube Creation using Wizard....7
2.4 Dimension Design....8
2.5 Cube Structure Design....10
2.6 Browsing the Cube....10
2.7 Deployment and Processing....11
3 Demonstration of OLAP operations....12
3.1 Connecting Excel to SSAS Cube....12
3.2 Creating Pivot Table....13
3.3 OLAP Operations....13
3.3.1 Roll up....13
3.3.4 Drill down....14
3.3.6 Dice....15
3.3.7 Pivot (Rotate view)....15
3.4 Conclusion....16
4 Power - BI Reports....16
4.1 Data Preparation and Connection....16
4.2 Data Modeling....16
4.3 Report 1: Matrix Visual....17
4.4 Report 2: Slicers with Multiple Visuals....18
4.5 Report 3: Drill Down Report....19
4.6 Report 4: Drill Through Report....20
4.7 Conclusion....20

1 Data source

1.1 Data Source

This dataset captures financial transaction data related to money transfers between accounts across different locations, currencies, and payment methods. It is modeled after a real world OLTP (Online Transaction Processing) system, where daily banking transactions are recorded and monitored.

The dataset includes transactional details such as sender and receiver accounts, transaction amounts, currencies, payment types, and geographic locations. Additionally, it contains indicators related to suspicious or laundering activities, making it suitable for analytical processing.

The raw transactional data is first loaded into a staging database, where data cleaning and transformation are performed. The processed data is then loaded into the data warehouse, where it is structured into fact and dimension tables to support analytical queries and reporting.

1.2Entities and Their Purpose

1. DimAccount

- Stores information about accounts involved in transactions.
- Used as a role-playing dimension (Sender Account & Receiver Account).

Attributes:

- AccountKey
- AccountID

2. DimLocation

- Stores geographic information related to bank locations.
- Used as a role-playing dimension (Sender Location & Receiver Location).

Attributes:

- LocationKey
- LocationName

3. DimCurrency

- Contains details about currencies used in transactions.

Attributes:

- CurrencyKey
- CurrencyCode

4. DimPaymentType

- Stores different types of payment methods used in transactions.

Attributes:

- PaymentTypeKey
- PaymentTypeName

5. DimDate

- Stores date-related information for transaction analysis.
- Supports time-based analysis such as year, month, and day trends.

Attributes:

- DateKey
- FullDate
- Year
- Quarter
- Month
- Day

6. FactTransactions

- Represents the core transactional data.
- Each record corresponds to a financial transaction between two accounts.
- Measures and indicators are used for analytical processing.

Measures / Attributes:

- TransactionID
- DateKey
- SenderAccountKey

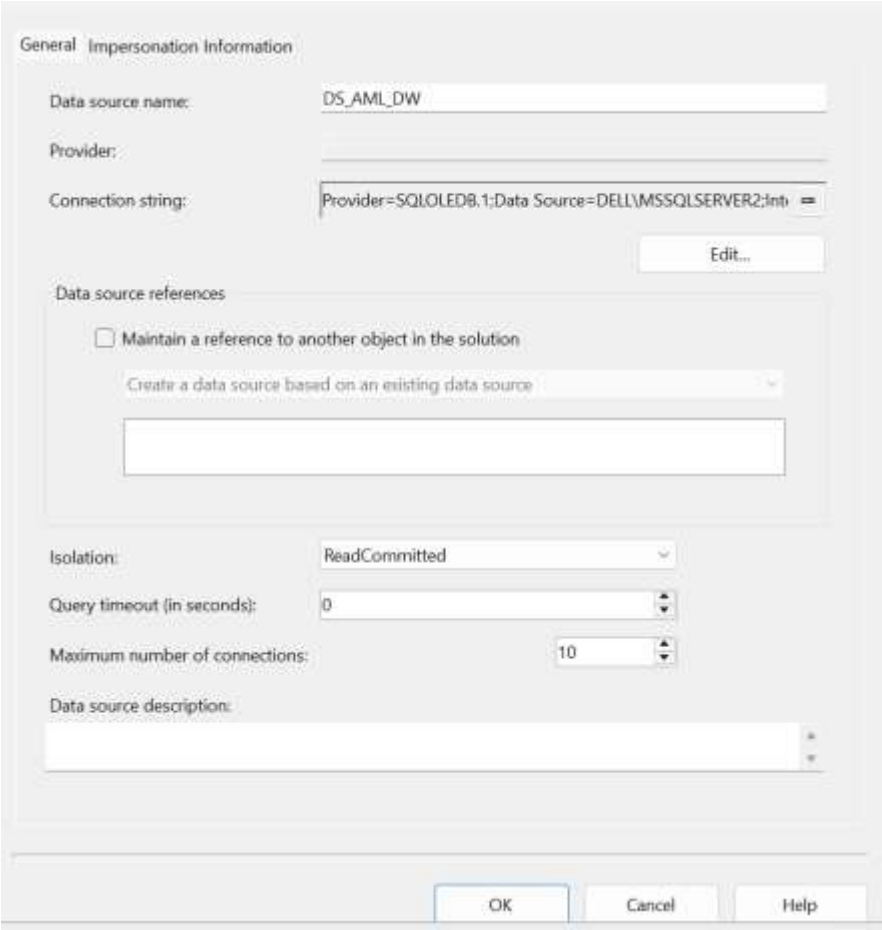
- ReceiverAccountKey
- SenderLocationKey
- ReceiverLocationKey
- CurrencyKey
- PaymentTypeKey
- Amount
- Is_Laundering
- Is_Suspicious

2 SSAS Cube implementation

2.1 Data Loading into Data Source

First, the dataset from the data warehouse (AML_DW) was connected to SQL Server Analysis Services (SSAS) using the Data Source Designer.

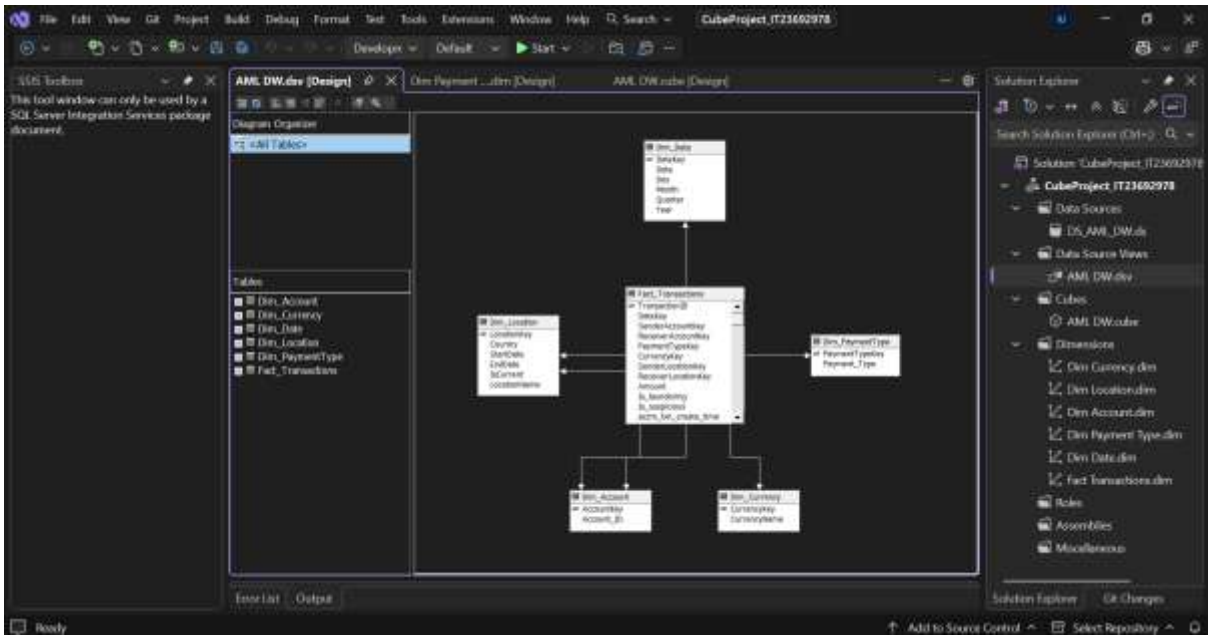
- A new Data Source was created.
- The connection was configured using:
    - Server name: localhost\SSAD
- Windows Authentication was used to establish the connection.



2.2 Creating Data Source View (DSV)

Next, a Data Source View (DSV) was created to define the logical schema for the cube.

- Required tables were selected from the data warehouse:
  - Fact table: Fact_Transactions
  - Dimension tables:
    - Dim_Date
    - Dim_Currency
    - Dim_Location
    - Dim_PaymentType
    - Dim_Account
- Relationships between fact and dimension tables were automatically detected or manually adjusted.

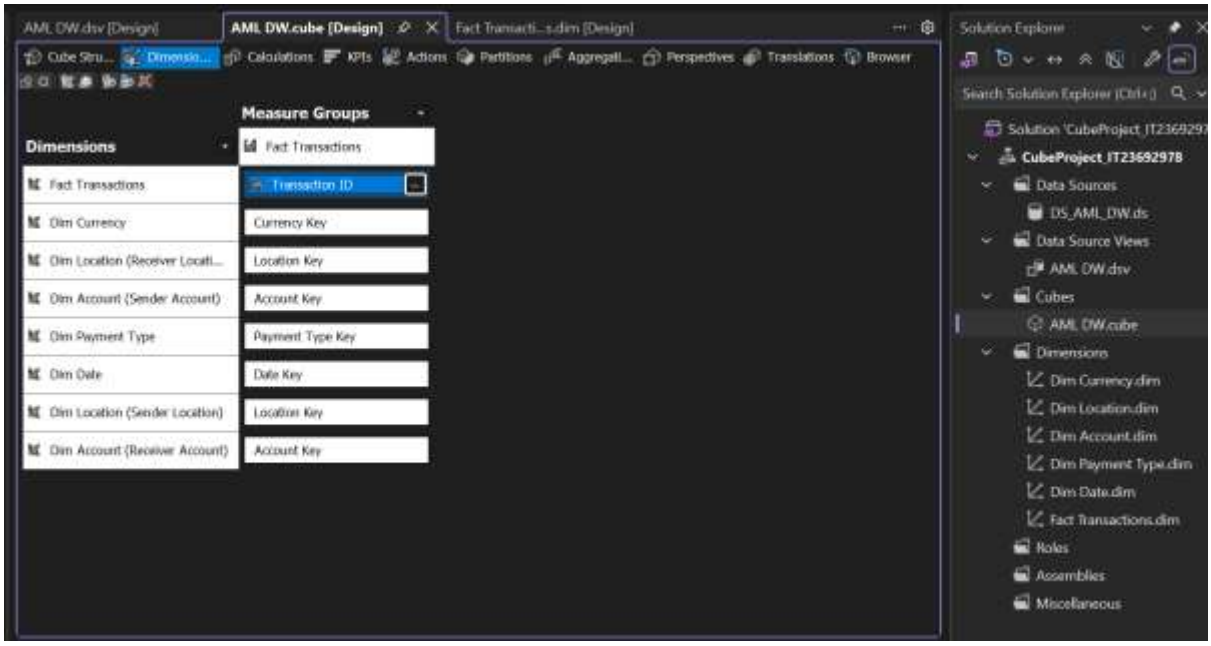


2.3 Cube Creation using Wizard

The cube was created using the Cube Wizard in SSAS.

Steps followed:

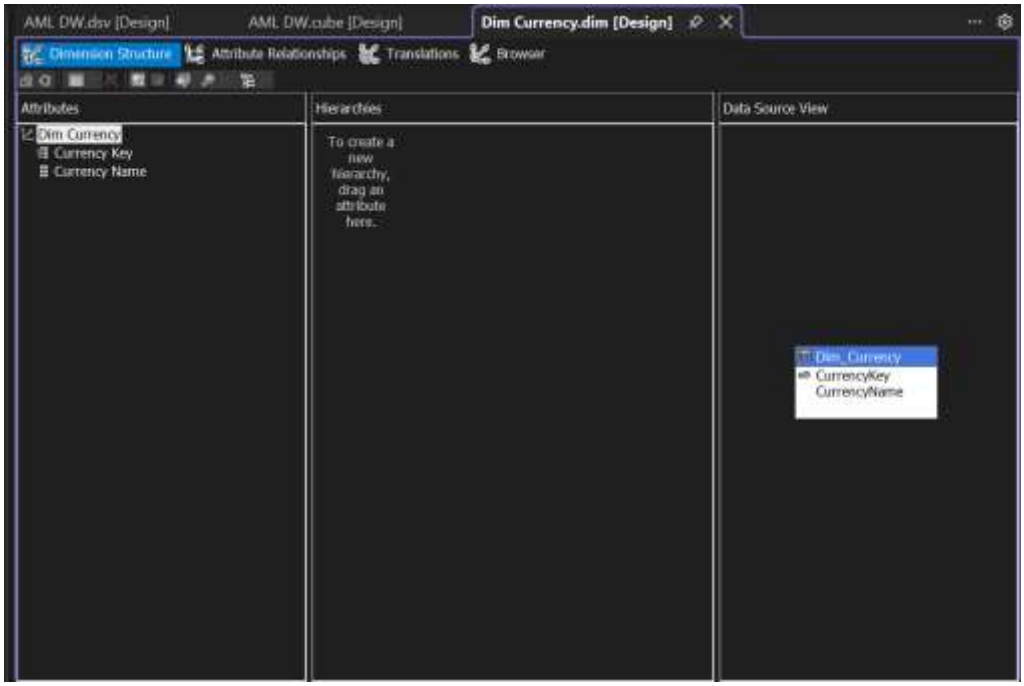
1. Selected Use existing tables option.
2. Chose the Fact_Transactions table as the main fact table.
3. Selected relevant measures:
    - Amount
    - Fact Transactions Count
    - Transaction Process Time Hours
4. Selected dimension tables:
    - Dim_Date
    - Dim_Currency
    - Dim_Location
    - Dim_PaymentType
    - Dim_Account

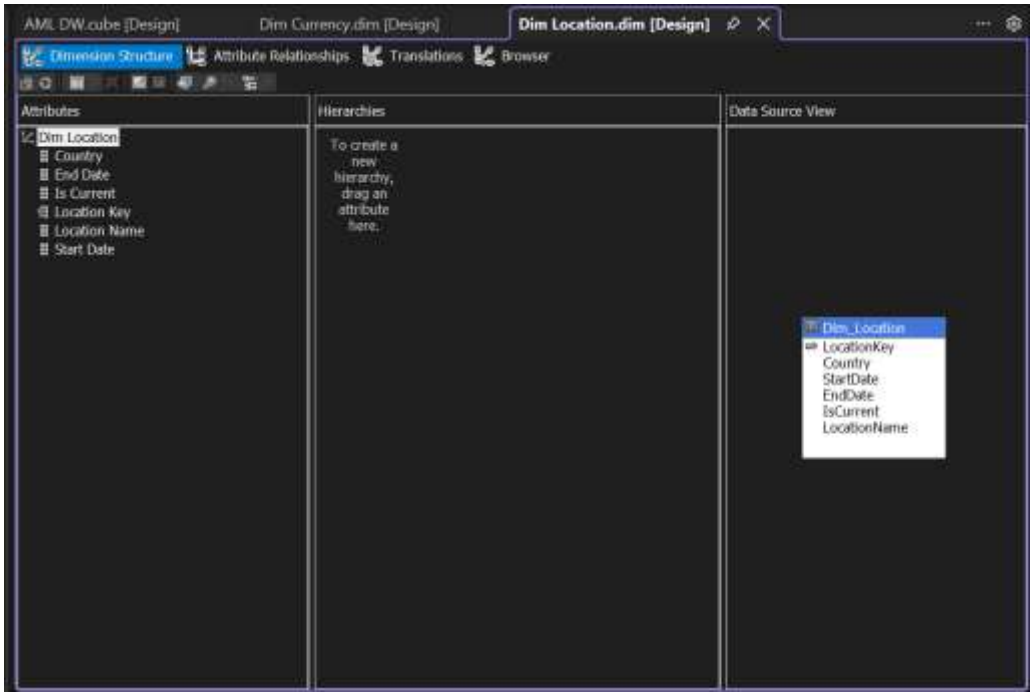


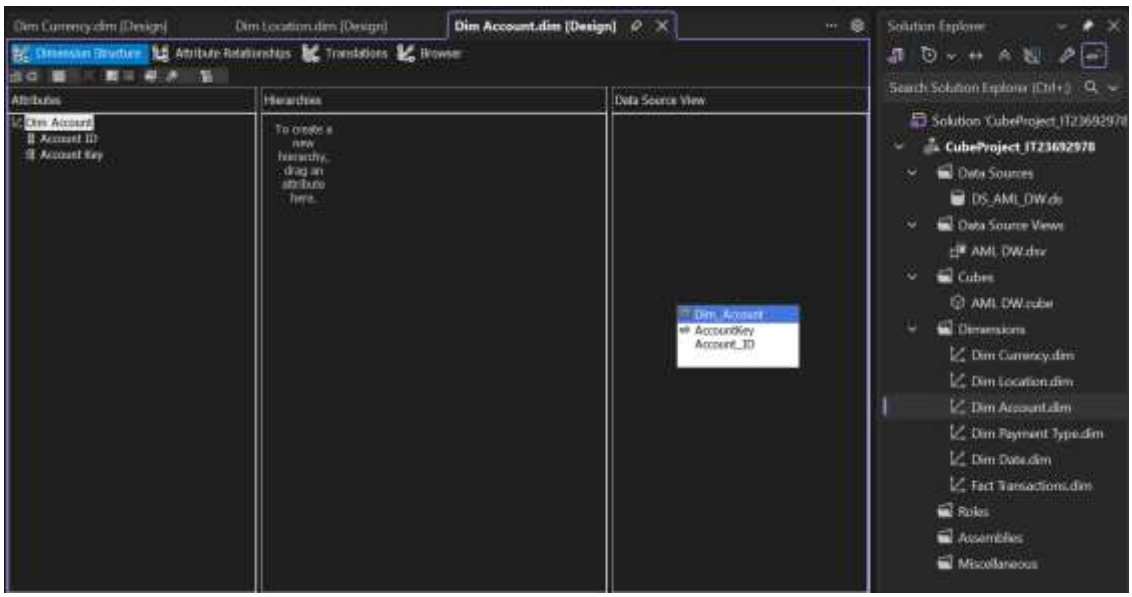
2.4 Dimension Design

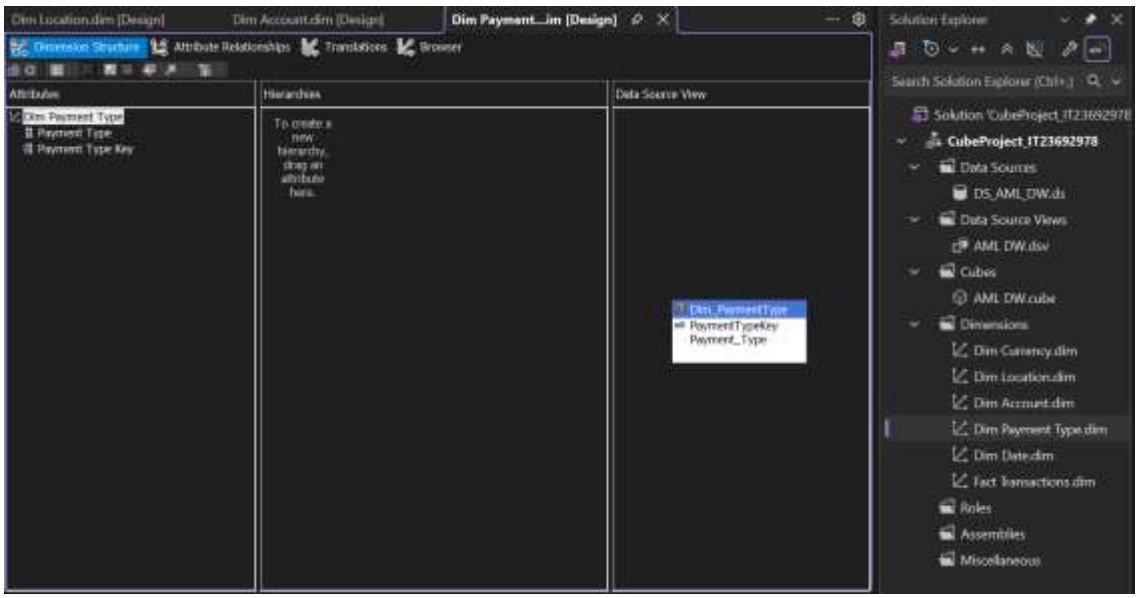
Dimensions were created and refined:

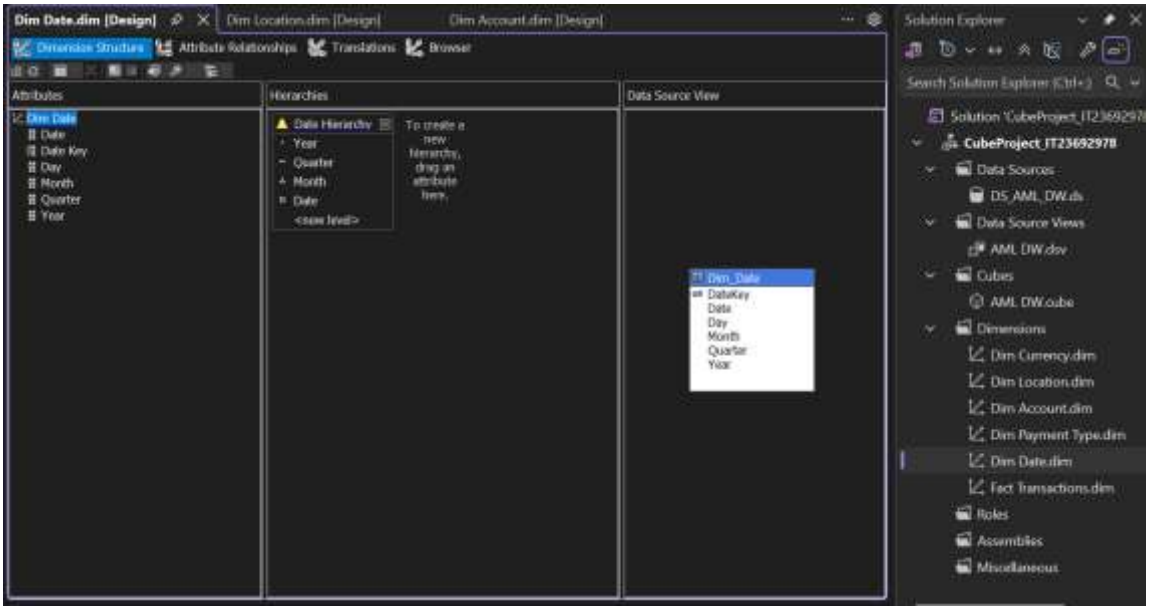
- Dim Date → Year, Quarter, Month, Day
- Dim Currency → Currency Name
- Dim Location → Country, Location Name
- Dim Payment Type → Payment Type
- Dim Account → Account ID






Dim Currency.dim [Design]
Dim Location.dim [Design]
Dim Account.dim [Design]
Dimension Structure
Attribute Relationships
Translations
Browser
Attributes
Hierarchies
Data Source View
Dim Account
Account ID
Account Key
To create a new hierarchy, drag an attribute here.
Dim_Account
AccountKey
Account_ID
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'CubeProject_IT23692978'
CubeProject_IT23692978
Data Sources
DS_AML_DW.ds
Data Source Views
AML_DW.dsv
Cubes
AML_DW.cube
Dimensions
Dim Currency.dim
Dim Location.dim
Dim Account.dim
Dim Payment Type.dim
Dim Date.dim
Fact Transactions.dim
Roles
Assemblies
Miscellaneous

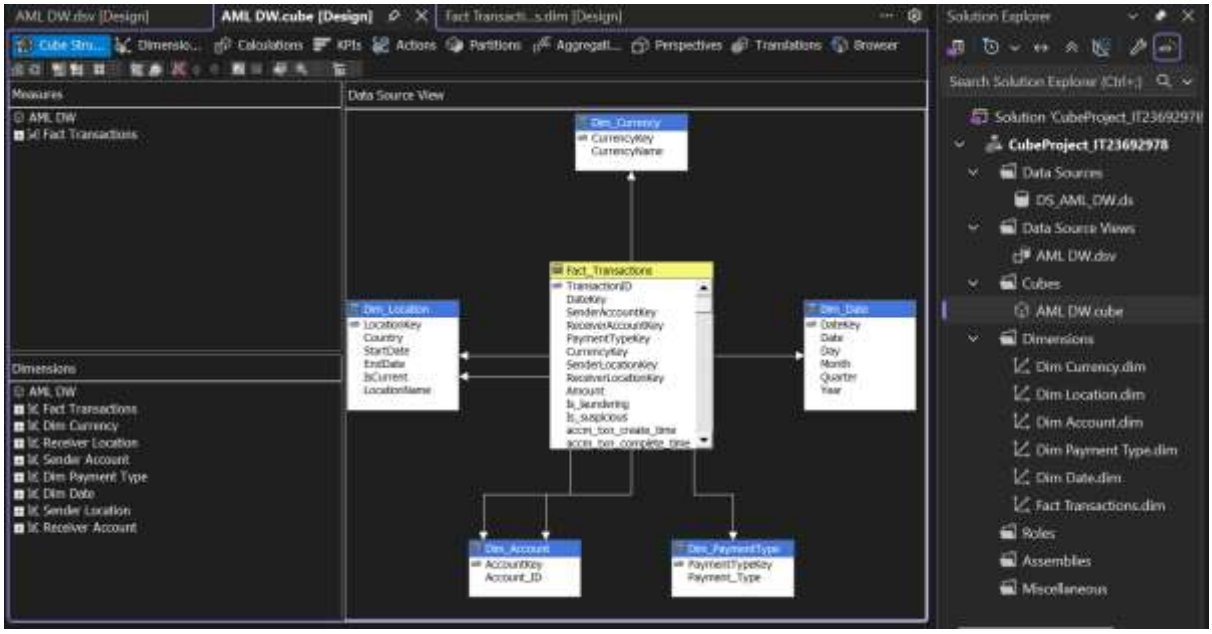

Dim Location.dim [Design]
Dim Account.dim [Design]
Dim Payment...im [Design]
Dimension Structure
Attribute Relationships
Translations
Browser
Attributes
Hierarchies
Data Source View
Dim Payment Type
Payment Type
Payment Type Key
To create a new hierarchy, drag an attribute here.
Dim_PaymentType
PaymentTypeKey
Payment_Type
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'CubeProject_IT23692978'
CubeProject_IT23692978
Data Sources
DS_AML_DW.ds
Data Source Views
AML_DW.dsv
Cubes
AML_DW.cube
Dimensions
Dim Currency.dim
Dim Location.dim
Dim Account.dim
Dim Payment Type.dim
Dim Date.dim
Fact Transactions.dim
Roles
Assemblies
Miscellaneous


Dim Date.dim [Design]
Dim Location.dim [Design]
Dim Account.dim [Design]
Dimension Structure
Attribute Relationships
Translations
Browser
Attributes
Hierarchies
Data Source View
Dim Date
Date
Date Key
Day
Month
Quarter
Year
Date Hierarchy
Year
Quarter
Month
Date
<new level>
To create a new hierarchy, drag an attribute here.
Dim_Date
DateKey
Date
Day
Month
Quarter
Year
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'CubeProject_IT23692978'
CubeProject_IT23692978
Data Sources
DS_AML_DW.ds
Data Source Views
AML_DW.dsv
Cubes
AML_DW.cube
Dimensions
Dim Currency.dim
Dim Location.dim
Dim Account.dim
Dim Payment Type.dim
Dim Date.dim
Fact Transactions.dim
Roles
Assemblies
Miscellaneous

2.5 Cube Structure Design

After the wizard, the cube was customized:

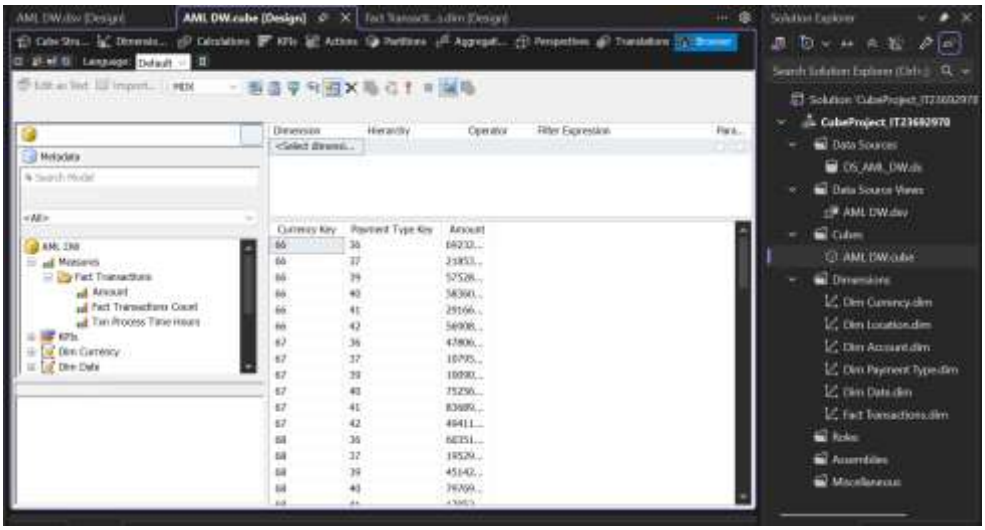
- Measures were organized in measure groups
- Dimensions were linked to the fact table
- Attribute relationships were defined for performance



2.6 Browsing the Cube

After processing, the cube was tested:

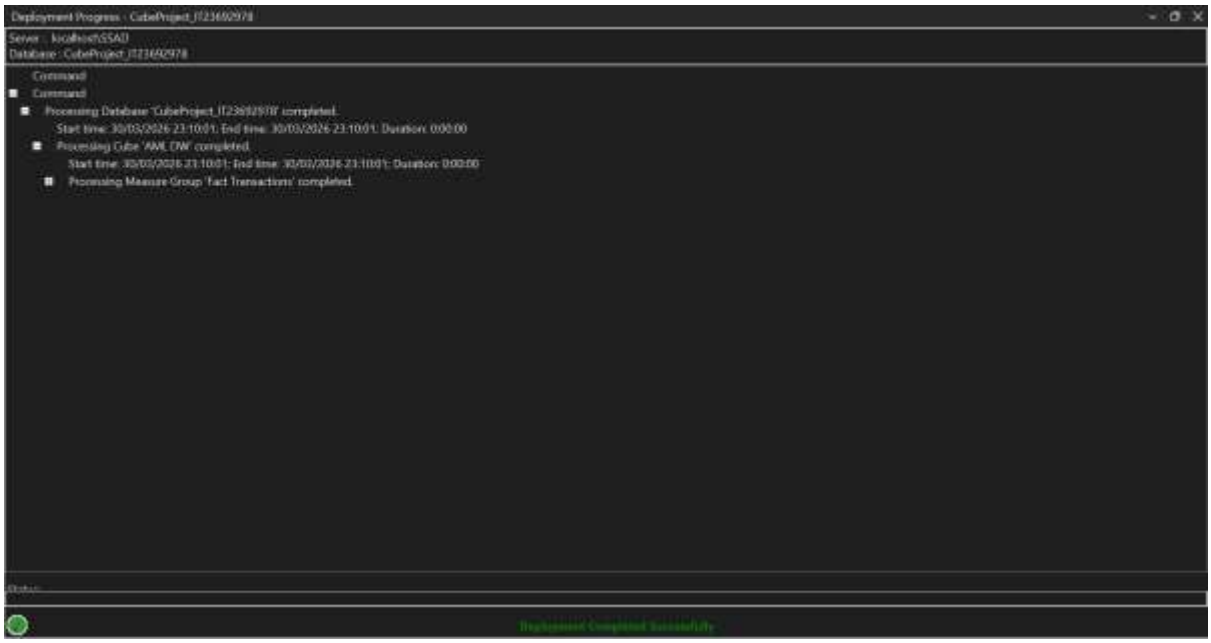
- Used the Browser tab in SSAS
- Dragged dimensions and measures to verify results
- Checked aggregations such as:
    - Total Amount by Currency



2.7 Deployment and Processing

The cube was then deployed and processed:

- The project was deployed to the SSAS server (localhost)
- Data was processed from the data warehouse into the cube
- All dimensions and measures were successfully processed



3 Demonstration of OLAP operations

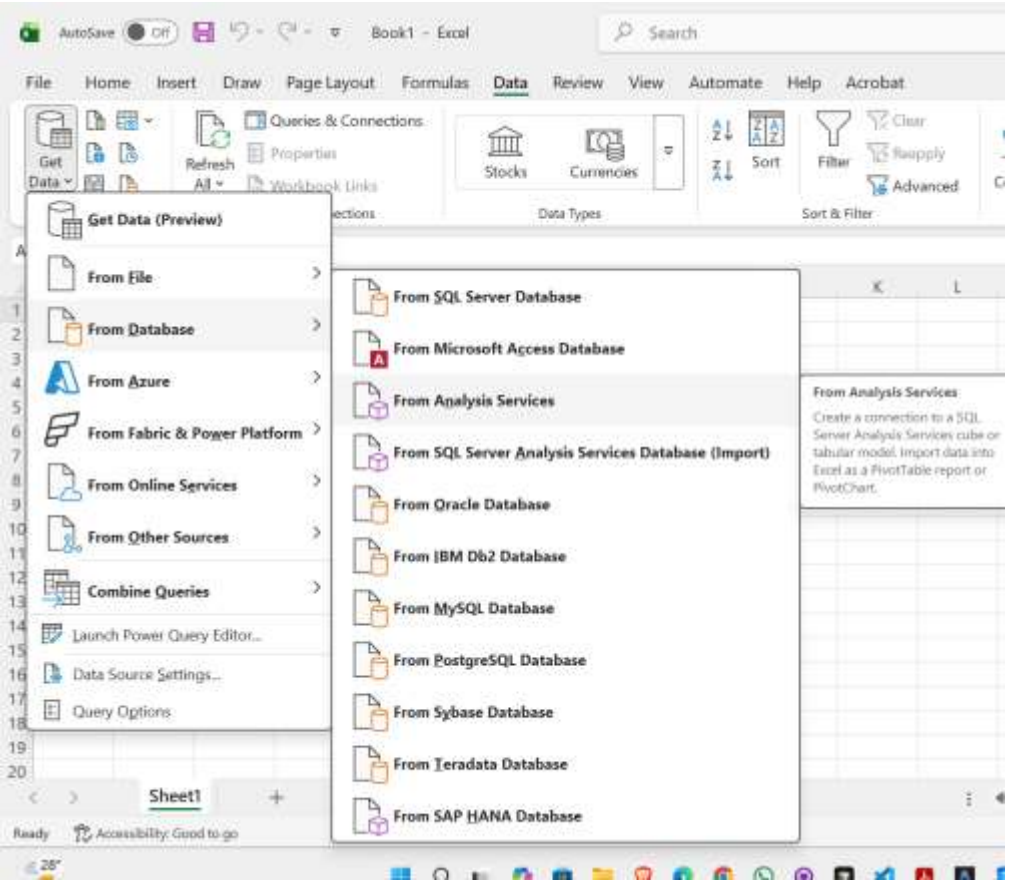
3.1 Connecting Excel to SSAS Cube

1. Connecting Excel to SSAS Cube

First, Excel was connected to the SSAS Cube.

Steps:

- Opened Microsoft Excel
- Went to Data tab → Get Data → From Cube → From Analysis Services
- Entered server name: localhost/SSAD
- Selected the cube database (AML_DW)
- Clicked Load → PivotTable

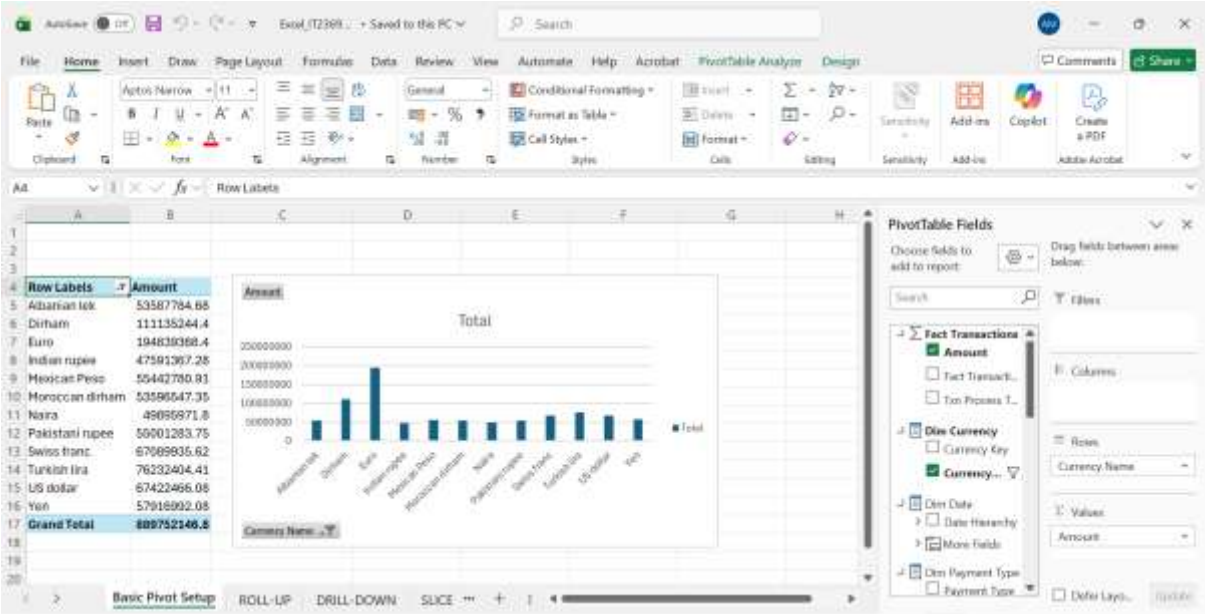


3.2 Creating Pivot Table

A Pivot Table was created using cube fields.

Fields used:

- Rows → Currency Name / Date
- Columns → Payment Type



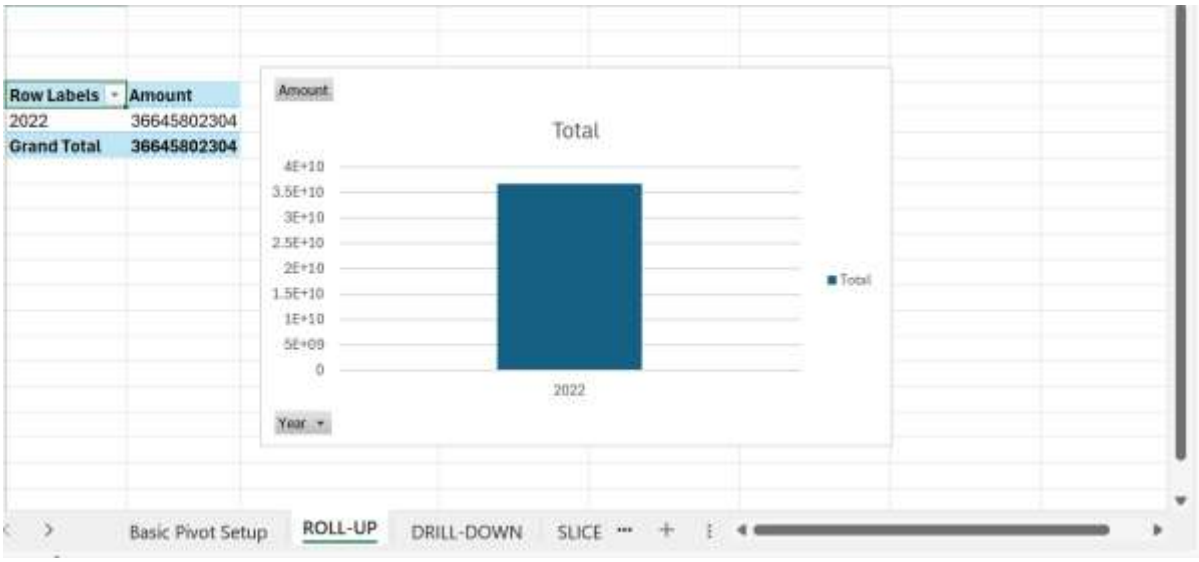
3.3 OLAP Operations

3.3.1 Roll up

Roll up is used to summarize data at a higher level.

Example:

- Used Date hierarchy
- Viewed total Amount by Year

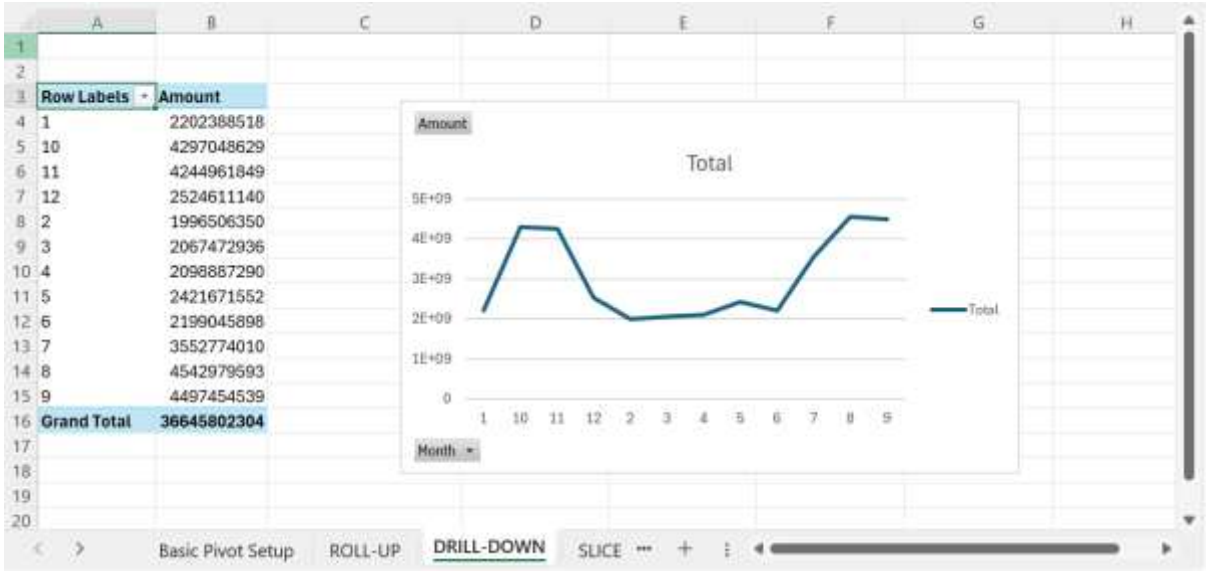


3.3.4 Drill down

Drill down is used to view detailed data.

Example:

- Expanded Year → Month
- Observed monthly transaction values



Row Labels	Amount
1	2202388518
10	4297048629
11	4244961849
12	2524611140
2	1996506350
3	2067472936
4	2098887290
5	2421671552
6	2199045898
7	3552774010
8	4542979593
9	4497454539
Grand Total	36645802304

3.3.5 Slice (Single filter)

Slice is used to filter data using one dimension.

Example:

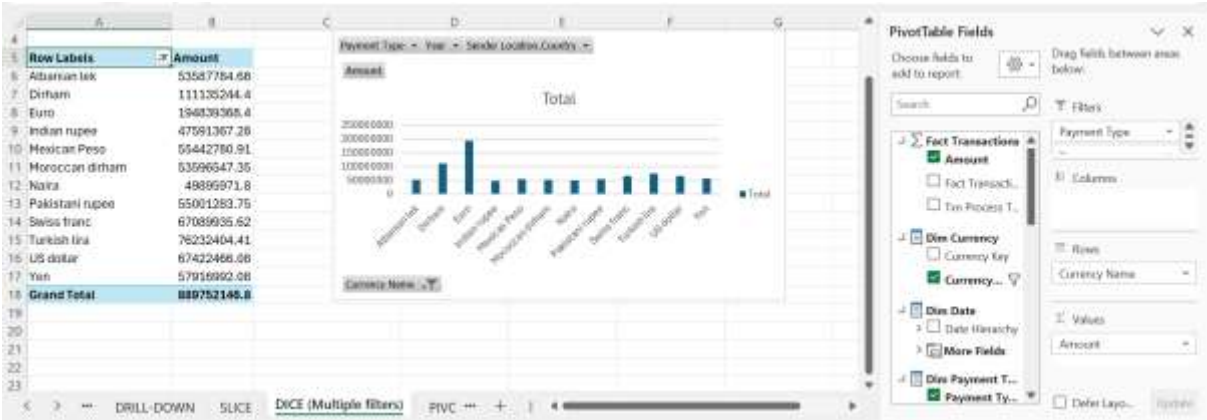
- Applied filter on Currency Name = USD
- Viewed only USD transactions

3.3.6 Dice

Dice is used to filter data using multiple dimensions.

Example:

- Applied filters:
  - Currency = EURO
  - Payment Type = Credit Card
  - Year = 2022

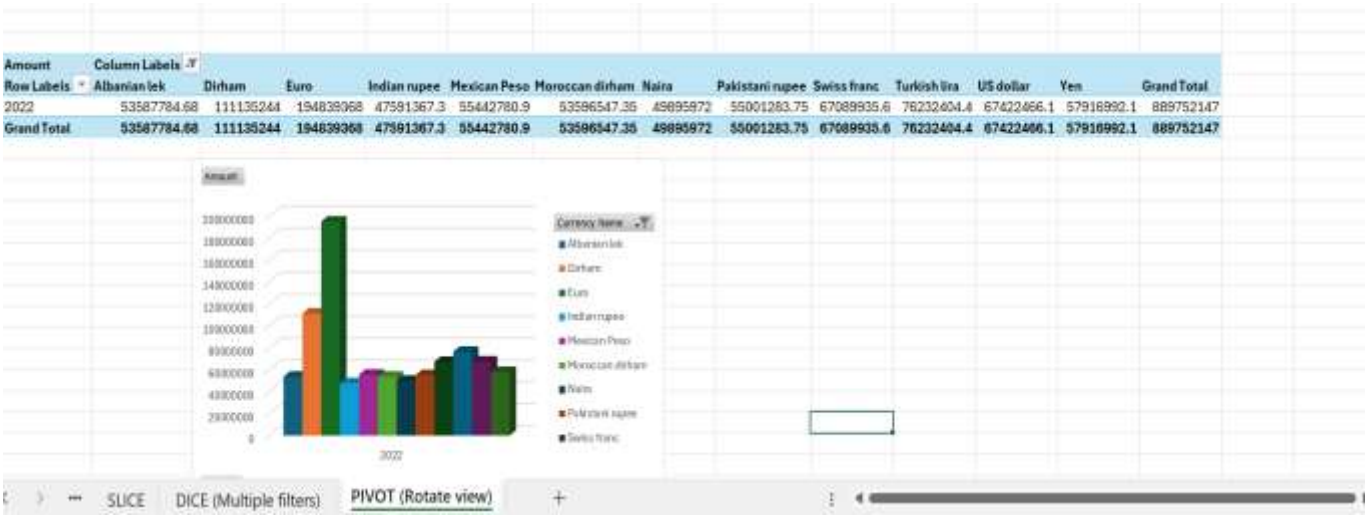


3.3.7 Pivot (Rotate view)

Pivot is used to change the orientation of the data.

Example:

- Swapped Rows and Columns:
  - Currency → Columns
  - Payment Type → Rows



Amount	Column Labels												
Row Labels	Albanian lek	Dirham	Euro	Indian rupee	Mexican Peso	Moroccan dirham	Naira	Pakistani rupee	Swiss franc	Turkish lira	US dollar	Yen	Grand Total
2022	53587784.68	111135244	194839368	47591367.3	55442780.9	53596547.35	49895972	55001283.75	67089935.6	76232404.4	67422466.1	57916992.1	889752147
Grand Total	53587784.68	111135244	194839368	47591367.3	55442780.9	53596547.35	49895972	55001283.75	67089935.6	76232404.4	67422466.1	57916992.1	889752147

3.4 Conclusion

Excel Pivot Tables were used to demonstrate OLAP operations such as roll up, drill down, slice, dice, and pivot. These operations allow users to analyze multidimensional data effectively from different perspectives.

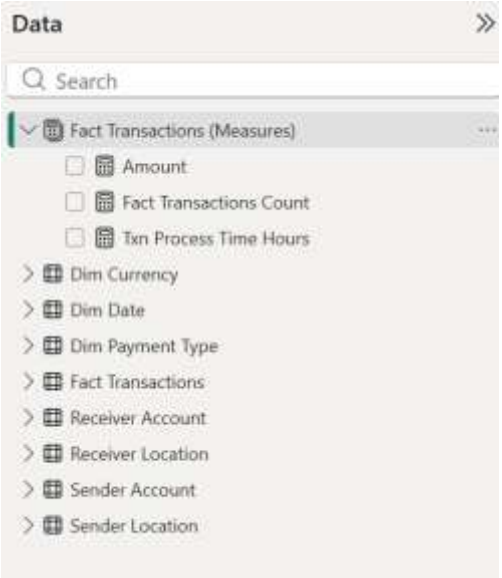
4 Power - BI Reports

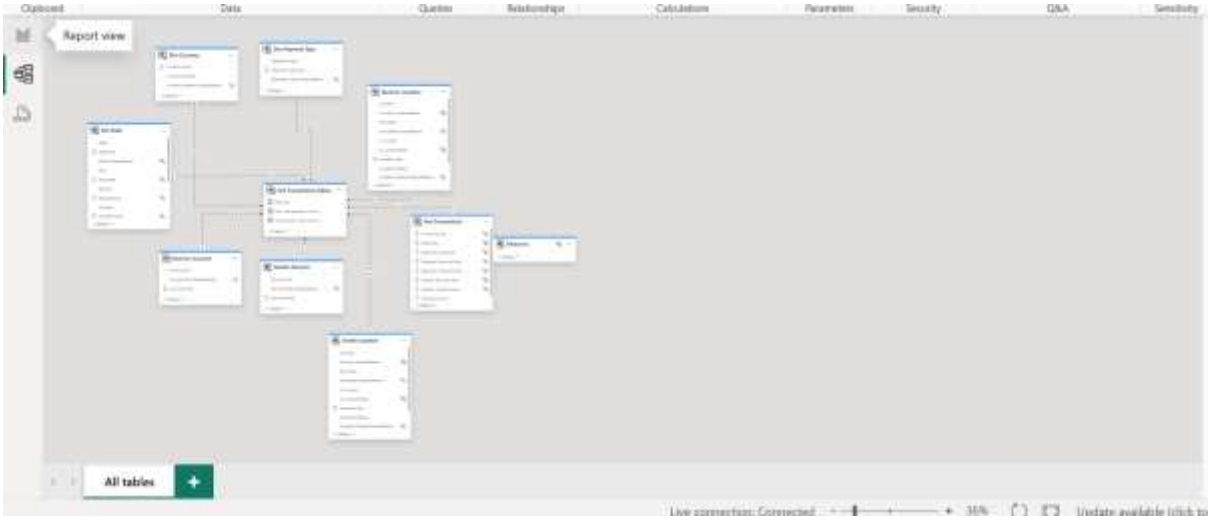
4.1 Data Preparation and Connection

- Opened Power BI Desktop
- Connected to SSAS Cube (Live Connection)
- Selected cube database (AML_DW)
- Loaded available dimensions and measures

4.2 Data Modeling

- Used existing cube model (no additional relationships needed)
- Used cube measures:
  - Amount
  - Fact Transactions Count
- Used dimension fields:
  - Currency Name
  - Payment Type
  - Country
  - Date Hierarchy



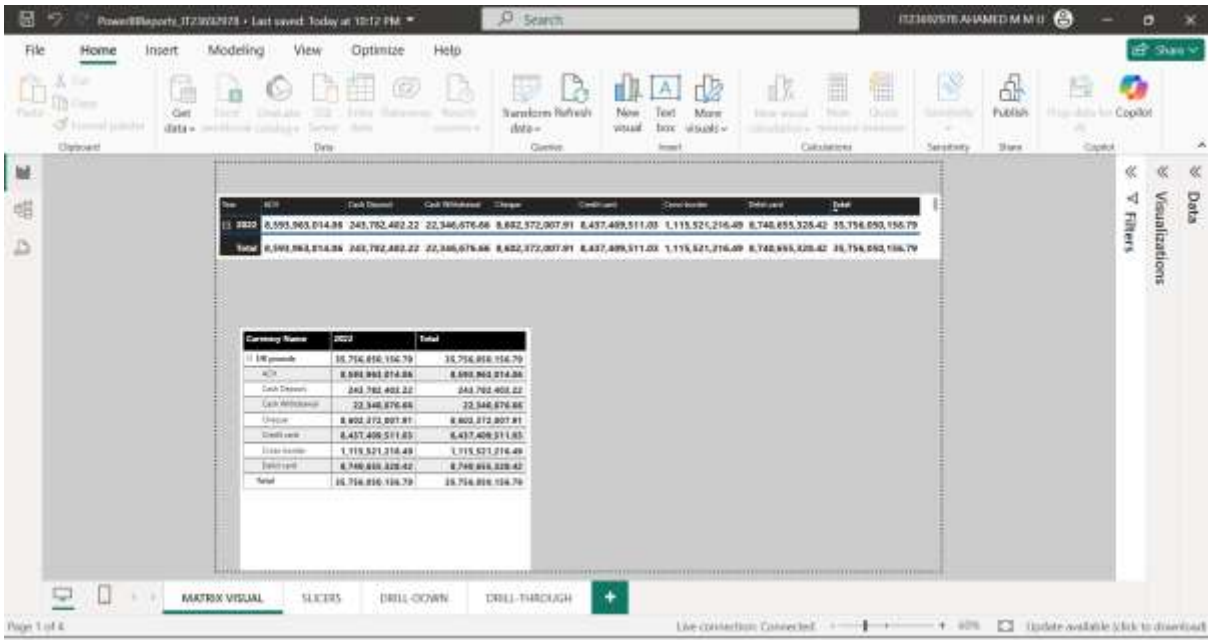


4.3 Report 1: Matrix Visual

To display detailed tabular data with row and column groupings

A Matrix visual was created to present detailed tabular data. Currency Name was placed in Rows, Year in Columns, and Total Amount in Values. This allowed row and column grouping for multidimensional analysis.

- Inserted Matrix visual
- Dragged fields:
  - Rows → Currency Name
  - Columns → Payment Type
  - Values → Amount

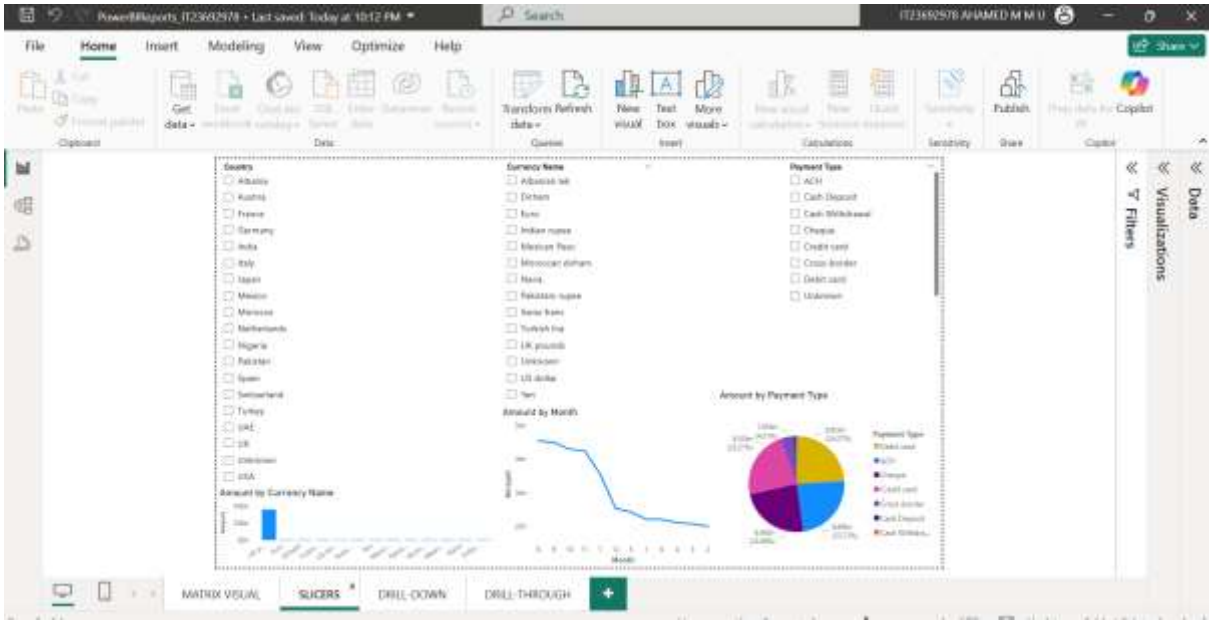


4.4 Report 2: Slicers with Multiple Visuals

To create an interactive report with multiple slicers and multiple visuals.

Three slicers were created using Country, Currency Name, and Payment Type. Each slicer was built by inserting a slicer visual and placing the corresponding field into the slicer field well. Cascading filtering was achieved through the data model relationships and visual interactions. Additional visuals including a bar chart, pie chart, and line chart were used to present insights interactively.

- Added slicers:
  - Slicer 1 → Country
  - Slicer 2 → Currency Name
  - Slicer 3 → Payment Type
- Added visuals:
  - Bar chart → Currency Name vs Amount
  - Line chart → Month vs Amount
  - Pie chart → Payment Type vs Amount

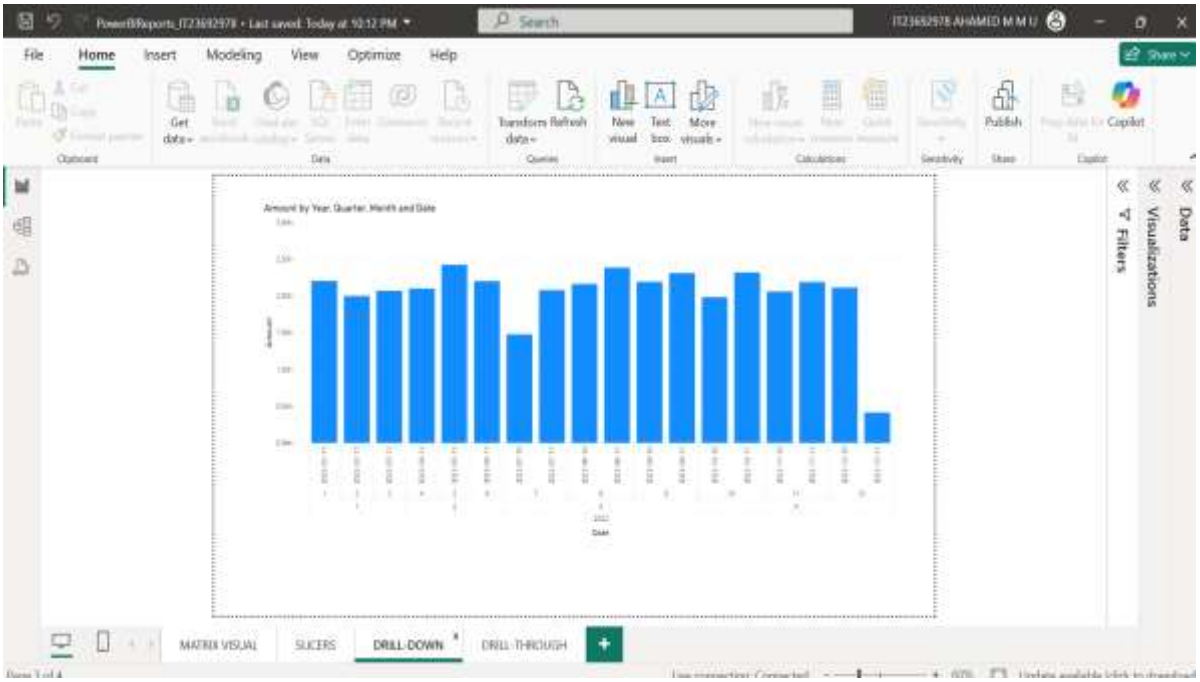


4.5 Report 3: Drill Down Report

To allow users to explore data hierarchically from Year to Quarter to Month

A drill down report was created using a chart with a Date Hierarchy. The hierarchy contained Year, Quarter, and Month levels. Total Amount was used as the measure. This allowed users to move from summary level data to more detailed time based analysis.

- Inserted Column chart
- Dragged fields:
    - X axis → Date Hierarchy (Year → Quarter → Month)
    - Y axis → Amount
- Enabled drill down option



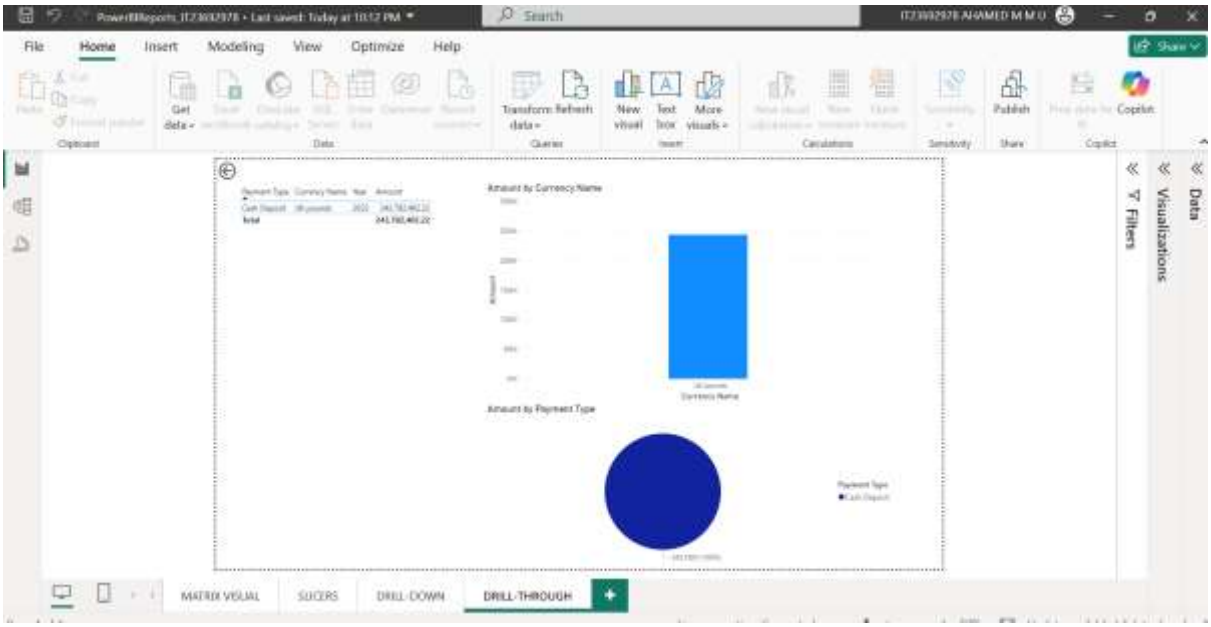
4.6 Report 4: Drill Through Report

To allow users to right click a visual and navigate to a detailed page.

A drill through report was created by adding a detail page and assigning Currency Name as the drill through field. A summary visual on the main page allowed users to right click on a selected item and navigate to a filtered detail page containing transaction level information.

Drill-through lets the user:

- Right click a value in one visual
- open another page
- see detailed information filtered for that selected value



4.7 Conclusion

Power BI reports were created using data from the SSAS cube. The reports demonstrate matrix visualization, slicer based filtering, drill down navigation, and drill through analysis, enabling interactive and meaningful data exploration.